

Module-3.

Instance based Learning Algorithms:- Smaller

time taken for training is long

• Lazy learners

• Processing of training instances is done only during testing phase

• No model is built with the training instances before it receives a test instance.

• Predicts the class of the test instances directly from the training data.

• Slow in testing phase

• Learns by making many local approximations.

Model-based learning.

• Eager learners

• Processing of training instances is done during training phase.

• Generalizes a model with the training instances before it receives a test instance.

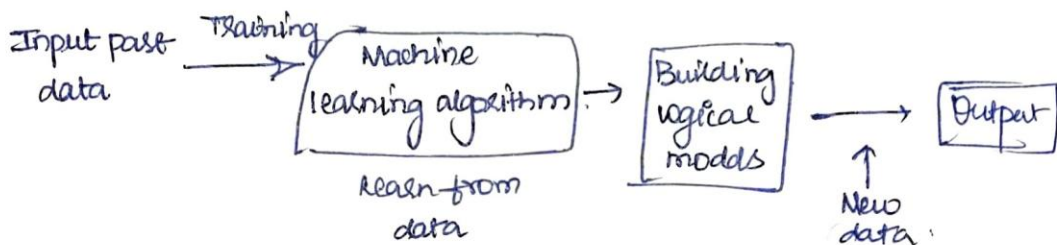
• Predicts the class of the test instance from the model built.

• Fast in testing phase.

• Learns by creating global approximation.

Similarity based learning:-

Model-based learning.



Instance based learning algorithms:-

- ① K-Nearest Neighbor (K-NN)
- ② Variants of nearest neighbor learning
- ③ Locally weighted regression
- ④ Learning vector quantization (LVQ)
- ⑤ Self-organizing map (SOM)
- ⑥ Radial Basis Function (RBF) networks

① K-Nearest Neighbor (K-NN)

↓
Represents the number of neighbors to be considered.

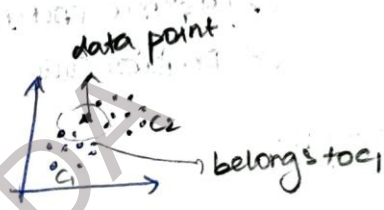
T → the sample to be selected.

↓
has to be found.

How it works?

- choose no. of neighbors. $K=5$ → Find Euclidean distance b/w data point based on K → The nearest point will be considered & data point is considered that
- calculate the Euclidean distance between the data points. The Euclidean distance is the distance b/w two points, which we have already studied and is calculated as.

$$\text{Euclidean distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$



Algorithm:-

Inputs: Training dataset T, distance metric d, Test instance t, the number of nearest neighbors K.

Output: Prediction class or category.

Prediction: For test instance t,

- ① For each instance i in T, compute the distance between the test instance t and every other instance i in the training data using a distance metric (Euclidean distance).
 - ② Sort the distances in an ascending order and select the first K nearest training data instances to the test instance.
 - ③ Predict the class of the test instance by majority voting or as mean of the K selected nearest instances.
- majority voting → if target attribute is discrete value.
 - mean → if target attribute is continuous value.

Q1- Consider the Student Performance training dataset of 8 data instances shown in Table which describes the performance of individual students in a course and their CGPA obtained in the previous semesters. The independent attributes are CGPA, Assessment and Project. The target variable is 'Result' which is a discrete valued variable that takes two values 'Pass' or 'Fail'. Based on the performance of a student, classify whether a student will pass or fail in that course. Given a test instance (6.1, 40, 5) assign a class/category.

Note: KNN classifier performance is affected by

① K no. of neighbors

② distance metric

③ Decision rule

Samples:

SNO.	CGPA	Assessment	Project Submitted	Result
1.	9.2	85	8	45.206 Pass
2.	8	80	7	40.095 Pass
3.	8.5	81	8	41.179 Pass
4.	6	45	5	5.00099 Fail
5.	6.5	50	4	10.0578 Fail
6.	8.2	72	7	32.1311 Pass
7.	8.8	38	5	2.0223 Fail
8.	8.9	91	9	51.2331 Pass

Solⁿ:

Step

• The test instance is (6.1, 40, 5)

• compute the Euclidean distance b/w training data & test data.

$$\Rightarrow (9.2 - 6.1)^2 + (85 - 40)^2 + (8 - 5)^2$$

Sort the ~~distances~~ distances in ascending order. $K=3 \rightarrow$ assumption

Index	Distance	Class
⑦	2.0022	Fail
④	5.0099	Fail
⑤	10.0578	Fail
⑥	32.1311	Pass
②	40.095	Pass
③	41.179	Pass
①	45.206	Pass
⑧	51.2331	Pass

Selected -

⑦	④	⑤
2.0022	5.0099	10.0578
Fail	Fail	Fail

Prediction.

The test instance given (6.1, 40.5) is predicted as fail based on majority voting.

Weighted K-Nearest Neighbor Algorithm:-

Inputs: Training dataset 'T', Distance metric 'd', Weighting Function $w(i)$, Test instance 't', the number of nearest neighbors 'K'.

Output: Predicted class or category.

Prediction: For test instance 't',

Step 1:- For every instance 'i' in training dataset 'T', compute the distance between the test instance 't' and every other instance 'i' using a distance metric.

Step 2:- Sort the distances in the ascending order and select the first 'K' nearest training data instances to the test instance.

Step 3:- Predict the class of the test instance by weighted voting technique (Weighting function $w(i)$) for the 'K' selected nearest instances:

- Compute the inverse of each distance of the 'K' selected nearest instances.
- Find the sum of the inverses.
- Compute the weight by dividing each ~~instance~~ inverse distance by the sum.
- Add the weights of same class.
- Predict the class by choosing the class with the maximum vote.

Example:-

Consider the training dataset given in the table. Use Weighted K-NN and determine the class for the test instance (7.6, 60, 8).

<u>SNO.</u>	<u>CGPA</u>	<u>Assessment</u>	<u>Project Submitted</u>	<u>Result</u>	<u>Eucleidian distance</u>
1	9.2	85	8	Pass	25.051
2	8	80	7	Pass	20.0289
3	8.5	81	8	Pass	21.0192
4	6	45	5	Fail	15.3805
5	6.5	50	4	Fail	10.8263
6	8.2	72	7	Fail Pass	12.0565
7	5.8	38	5	Fail	22.2764
8	8.9	91	9	Pass	31.0433

Let us assume $K=3$.

Soln:-

⑤
10.826
Fail

⑥
12.056
~~Fail~~
Pass

④
15.3805
Fail.

↑
Step 1

<u>SNO.</u>	<u>CGPA</u>	<u>Assessment</u>	<u>Project</u>	<u>Result</u>	<u>Eucleidian</u>
5	6.5	50	4	Fail	10.826
6	8.2	72	7	Pass	12.056
4	6	45	5	Fail	15.3805

<u>Sno</u>	<u>Eucleidian distance</u>	<u>Inverse distance</u>	<u>Class</u>
5	10.826	$\frac{1}{10.826} = 0.0923$	Fail
6	12.056	$\frac{1}{12.056} = 0.0829$	Pass
4	15.3805	$\frac{1}{15.3805} = 0.0650$	Fail

Find Sum of Inverse values:-

Sum = 0.2402

Weight = $\frac{\text{Inverse}}{\text{Sum}}$

<u>Sno</u>	<u>Eucleidian</u>	<u>Inverse</u>	<u>Weight</u>	<u>Class</u>
5	10.826	0.0923	0.384	Fail
6	12.056	0.0829	0.345	Pass
4	15.3805	0.0650	0.270	Fail

Adding weights of some class:-

class Fail:-

$$\rightarrow 0.384 + 0.270 = 0.654$$

class Pass

$$\rightarrow 0.345$$

maximum vote.

~~∴ instance~~ test instance (7.6, 60, 8) is predicted to be in class Fail.

Nearest centroid classifier:-

Inputs: Training dataset T , Distance metric d , Test instance t .

Output: Predicted class or category.

1. Compute the mean/centroid of each class
2. Compute the distance between the test instance and mean/centroid of each class (Euclidean distance).
3. Predict the class by choosing the class with the smaller distance.

Example:-

Consider the same data shown in Table with features x & y . The target classes are 'A' or 'B'. Predict the class for test instance (6,5) using Nearest centroid classifier.

x	y	class
3	1	A
5	2	A
4	3	A
7	6	B
6	7	B
8	5	B

~~$$\text{mean } \bar{x} = \frac{3+5+4+7+8+6}{6} = 5.5$$~~

~~$$\text{mean } \bar{y} = \frac{1+2+3+6+7+5}{6} = 4$$~~

mean of class A

$$\bar{x} = \frac{3+5+4}{3} = 4$$

$$\bar{y} = \frac{1+2+3}{3} = 2$$

(4,2) $\rightarrow$ A

mean of class B

$$\bar{x} = \frac{7+6+8}{3} = 7$$

$$\bar{y} = \frac{6+7+5}{3} = 6$$

(7,6) $\rightarrow$ B

Euclidean distance b/w test & Centroid.

A
(8,5), (4,2)

B
(7,6), (6,5)

$$\Rightarrow \sqrt{(6-4)^2 + (5-2)^2} = 3.605$$

$$\sqrt{(7-6)^2 + (6-5)^2} = 1.41$$

Smaller distance class is B with 1.41

$\therefore$ The instance (6,5) belongs to class B.

Q:- Apply Find-S algorithm and candidate algorithm.
 $\hookrightarrow$ enjoy sport dataset

Example	Sky	Airtemp	Humidity	wind	water	forecast	Enjoy sport
1	Sunny	warm	Normal	Strong	warm	same	Yes
2	Sunny	warm	High	Strong	warm	same	Yes
3	Rain	cold	High	Strong	warm	change	No
4	Sunny	warm	High	Strong	cool	change	Yes

Find-S Algorithm:

$\langle \phi \ \phi \ \phi \ \phi \ \phi \ \phi \rangle$

$I_1 = \langle \text{Sunny warm Normal Strong warm same} \rangle \rightarrow \text{positive}$

$h_1 = \langle \text{Sunny warm Normal Strong warm same} \rangle$

$I_2 = \langle \text{Sunny warm High Strong warm same} \rangle \rightarrow \text{positive}$

$h_2 = \langle \text{Sunny warm ? Strong warm same} \rangle$

$I_3 = \langle \text{Rain cold High Strong warm change} \rangle \rightarrow \text{negative instance}$

$h_3 = \langle \text{Sunny warm ? Strong warm same} \rangle$

$I_4 = \langle \text{Sunny warm High Strong cool change} \rangle \rightarrow \text{positive}$

$h_4 = \langle \text{Sunny, warm ? Strong ? ?} \rangle$

Candidate elimination :-

step 1 :- $G = \langle ? ? ? ? ? ? \rangle$

step 2 :- $S = \langle \psi \psi \psi \psi \psi \psi \rangle$

$I_1 =$ Sunny warm normal strong warm same $\rightarrow$ positive instance.

$G_1 = \langle$ Sunny warm normal strong warm same $\rangle$

$G_1 = \langle ? ? ? ? ? ? \rangle$

$I_2 =$ Sunny warm high strong warm same $\rightarrow$ positive instance.

$S_2 = \langle$ Sunny warm ? strong warm same $\rangle$

$G_2 = \langle ? ? ? ? ? ? \rangle$

$I_3 =$ Rainy cold high strong warm change $\rightarrow$ Negative instance.

Specialize G_2 to exclude negative.

$\Rightarrow$ compare I_3 & S_2

$G_3 = \langle$ Sunny ? ? ? ? ? $\rangle$

$\langle ?$ warm ? ? ? ? $\rangle$

$\langle ? ? ? ? ?$ same $\rangle$

$S_3 = \langle$ Sunny warm ? strong warm same $\rangle$

$I_4 =$ Sunny warm high strong cool change $\rightarrow$ positive instance.

$S_4 = \langle$ Sunny warm ? strong ? ? $\rangle$

Prune G_3 to exclude all inconsistent with I_4 .

$G_4 = \langle$ Sunny ? ? ? ? ? $\rangle$

$\langle ?$ warm ? ? ? ? $\rangle$

version space :-

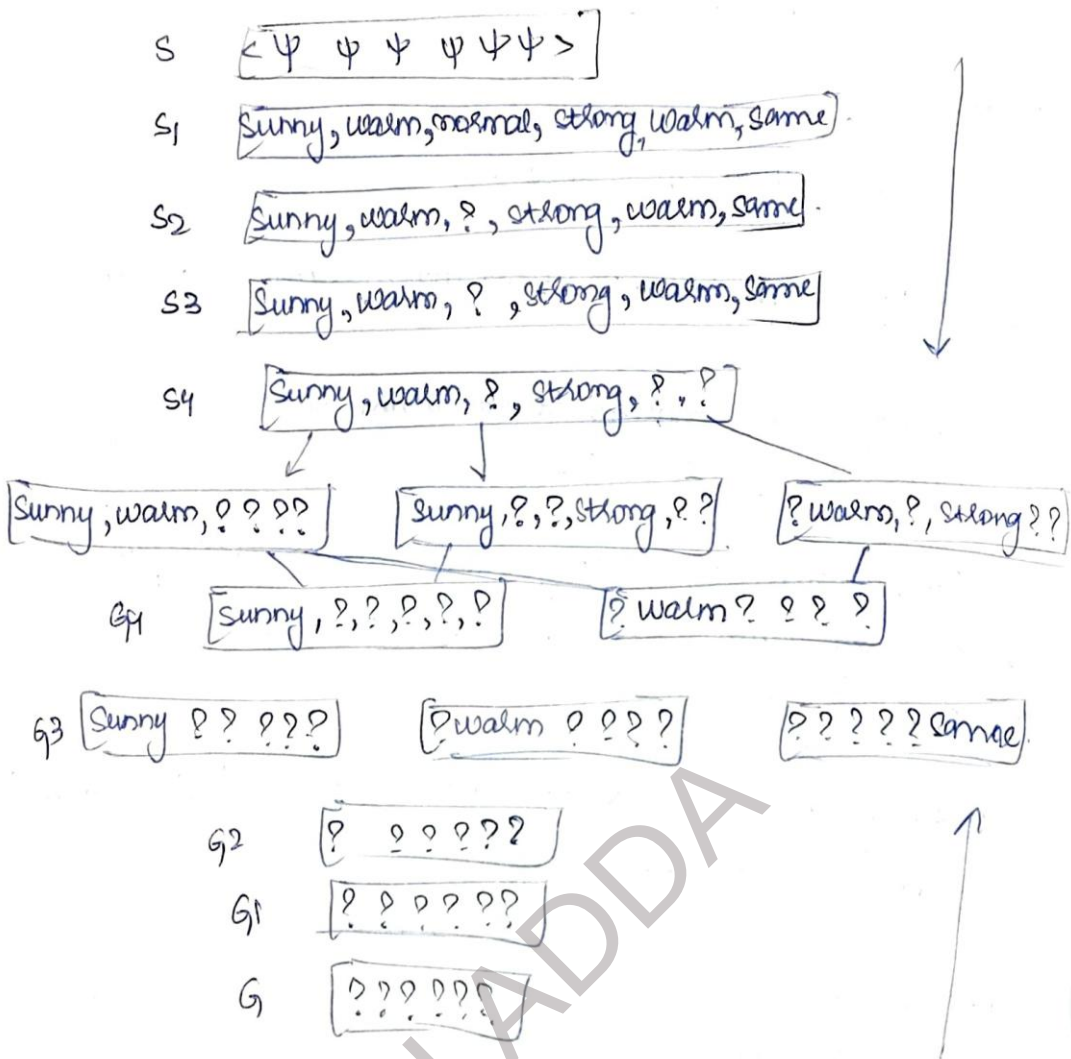
$S_1 = \langle$ Sunny warm ? strong ? ? $\rangle$

$\Rightarrow$

$\langle$ Sunny warm ? ? ? ? $\rangle$

$\langle$ Sunny ? ? strong ? ? $\rangle$

$\langle ?$ warm ? strong ? ? $\rangle$



Q-

Example	Eyes	Nose	Head	Color	Hair	Smile
1	Round	Triangle	Round	Purple	Yes	Yes
2	Square	Square	Square	Green	Yes	No
3	Square	Triangle	Round	Yellow	Yes	Yes
4	Round	Triangle	Round	Green	No	No
5	Square	Triangle Square	Round	Yellow	Yes	Yes

Find-S algorithm:-

$\langle \psi \ \psi \ \psi \ \psi \ \psi \ \psi \rangle$

$I_1 = \langle \text{Round Triangle Round Purple Yes} \rangle$ Positive instance.

$H_1 = \langle \text{Round Triangle Round Purple Yes} \rangle$

$I_2 = \langle \text{Square Square Square Green Yes} \rangle$ - Negative instance.

$h_2 = \langle \text{Round Triangle Round Purple Yes} \rangle$

I3 = < Square Triangle Round Yellow Yes > Positive instance.

h3 = < ? Triangle Round ? Yes >

I4 = < Round Triangle Round Green No > Negative instance.

h4 = < ? Triangle Round ? Yes >

I5 = < Square ~~Triangle~~ Round Yellow Yes > Positive instance.

h5 = < ? ~~Triangle~~ Round ? Yes >

Candidate Elimination:-

Step 1:-
S = < ? ? ? ? ? ? ? >
G = < ? ? ? ? ? ? ? >

Step 2:-
I1 = < Round Triangle Round Purple Yes > → Positive

S1 = < Round Triangle Round Purple Yes >

G1 = < ? ? ? ? ? ? >

I2 = < Square Square Square Green Yes > → Negative

~~G2 = < ? ? ? ? ? ? >~~

G2 = < Round ? ? ? ? ? >

< ? Triangle ? ? ? >

< ? ? Round ? ? >

< ? ? ? Purple ? >

S2 = < Round Triangle Round Purple Yes >

I3 = < Square Triangle Round Yellow Yes > Positive.

S3 = < ? Triangle Round ? Yes >

~~G3 = < Round ? ? ? ? >~~

< ? Triangle ? ? ? >

< ? ? Round ? ? >

$I_4 = \langle \text{Round Triangle Round Green No} \rangle$ Negative.

~~$G_4 = \langle ? \text{ Triangle } ? ? ? \rangle$~~

$\leftarrow$

$G_4 = \langle ? \text{ Triangle } ? ? ? \rangle$

$\langle ? ? ? ? \text{ Yes} \rangle$

$G_4 = \langle ? \text{ Triangle } ? ? ? \rangle$

$\langle ? \text{ Triangle } ? ? \text{ Yes} \rangle \checkmark$

~~$\leftarrow ? \text{ Triangle Round } ? ? \rightarrow$~~

$\langle ? ? \text{ Round } ? \text{ Yes} \rangle \checkmark$

$S_4 = ? \text{ Triangle Round } ? \text{ Yes}$

$S_5 = \langle \text{Square Square Round Yellow Yes} \rangle$ positive.

$S_5 = \langle ? ? \text{ Round } ? \text{ Yes} \rangle$

$G_5 = \langle ? ? \text{ Round } ? \text{ Yes} \rangle$

LWR:-

Regression:-

Linear Regression.

Logistic Regression.

- Linear regression is used for predicting continuous values. For example House price prediction, temperature prediction etc.
- Logistic regression is mainly used for classification problem. For example: spam detection, disease prediction, fraud detection etc.

LWR:-

LWR stands for Locally Weighted Regression.

- This is a non-parametric supervised learning algorithm which performs the local regression by combining the regression with nearest model.
- It requires training the data which prediction it uses only the training data instances locally around the point of interest. It

also uses nearest neighbor model where we find the instances that is close to the test instance and fit the linear function to each of those K -nearest instances in the local regression model.

It also finds the errors such that the prediction line is no more a linear but rather it is a curve.

$T \rightarrow$ training data.

$h_{\beta}(x) = \beta_0 + \beta_1 x$ $\rightarrow$ hypothesis function.
 β_0 is the intercept.
 β_1 is coefficient of x .

$J(\beta) = \frac{1}{2} \sum_{i=1}^m w_i (h_{\beta}(x_i) - y_i)^2$ $\rightarrow$ ②
 $\downarrow$
 called as cost function.

$w_i \rightarrow$ weight.
 $h_{\beta}(x) \rightarrow$ hypothesis function.
 $m \rightarrow$ no. of instances in training data sets/samples.
 $y \rightarrow$ True value.

Error function.
 $w_i = e^{-\frac{(x_i - x)^2}{\tau}}$ $\rightarrow$ ③
 $\uparrow$
 weight function — calculated by finding error.

$\tau \rightarrow (\text{tau})$ is called the bandwidth parameter.

The weight function used is Gaussian kernel for computation of the error.

τ which is a bandwidth parameter is used to control the rate at which w_i reduces to 0 with a distance from x_i .

① Consider the given training sample apply LWR for the same.

S.no.	Salary (x) (in lakhs)	Expenditure (y) (in thousands)
1	5	25
2	1	5
3	2	7
4	4	8

Let us assume that $\beta_0 = 4072$ $\beta_1 = 0.62$ & $\tau = 0.4$.

Let test instance $x = 2$.

Solⁿ:

Step 1: Given the test instance $x=2$

$$y = h_{\beta}(x) = \beta_0 + \beta_1 x$$

$$y = 4.72 + 0.62(2)$$

$$y = 5.96$$

Step 2: Apply nearest neighbor for the given training dataset.

Let us assume $k=3$

Sno	X	Eucledean distance
1	5	$\sqrt{(5-2)^2} = 3$
2	1	$\sqrt{(1-2)^2} = 1$
3	2	$\sqrt{(2-2)^2} = 0$
4	1	$\sqrt{(1-2)^2} = 1$

Solⁿ:

distance $\Rightarrow$

instance $\Rightarrow$ 3

1

2

1

4

$\therefore$ closer to test instance.

Step 3: The instances which are closer to the test instance is 2, 3, 4.

mean of y is

for instances 2, 3, 4

$$\mu = \frac{5+7+8}{3} = 6.67$$

$$\mu = 6.67$$

weight equation

$$w_i = e^{\frac{-(x_i - x)^2}{2\tau}}$$

weight for instances

$$w_2 = e^{\frac{-(x_2 - x)^2}{2\tau}} = e^{\frac{-(1-2)^2}{2 \times 0.4}} = 0.2865$$

$$w_3 = e^{\frac{-(2-2)^2}{2 \times 0.4}} = 1$$

$$w_4 = e^{\frac{-(1-2)^2}{2 \times 0.4}} = 0.2865$$

Step 4: compute hypothesis function for the test training data 2, 3, 4

$$y_i = h_{\beta}(x) = \beta_0 + \beta_1 x_i$$

$$y_2 = h_{\beta}(x_2) = \beta_0 + \beta_1(x_2)$$

$$y_2 = 4.72 + 0.62(1) = 5.34$$

$$h_{\beta}(2) = 4.72 + 0.62(2) = 5.96$$

$$h_{\beta}(1) = 4.72 + 0.62(1) = 5.34$$

steps:

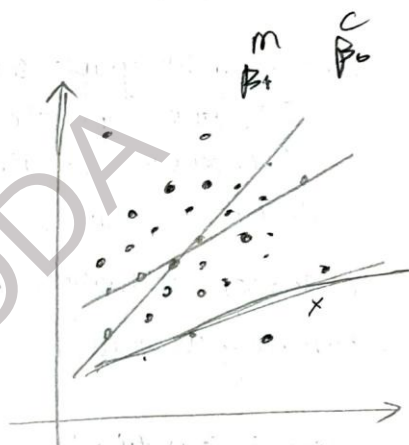
Cost function:-

$$J(\beta) = \frac{1}{2} \sum_{i=1}^m \{w_i(h_{\beta}(x_i) - y_i)^2\}$$

$$= \frac{1}{2} [0.2865 [5.34 - 5]^2 + 1 [5.96 - 7]^2 + 0.2865 [5.34 - 8]^2]$$

$$= 1.57$$

We need to adjust the cost function to minimize the error and get the optimal β parameters.



Regression analysis:-

It is a method of supervised learning. given a training data set D , sample N , training points x, y .

Regression analysis is mainly used to model the relationship between one or more independent variable x & dependent variable y .

The relationship b/w dependent & independent variable is defined by the function

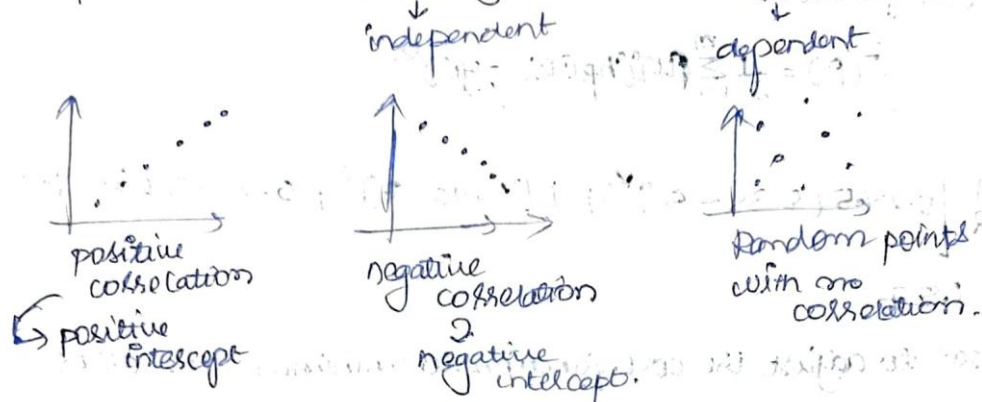
$$y = f(x) \text{ where } x \text{ is the feature variable}$$

$$y \text{ is the dependent variable.}$$

Introduction to Regression.

Regression & correlation - (2 marks).

Correlation among two variables can be done effectively using a scatter plot between explanatory variable & response variable.



- correlation by 2D graph showing relationship between 2 variables.
 - ① Positive correlation
 - ② Negative correlation.
 - ③ No correlation
- The Pearson correlation coefficient is the most common test for determining correlation if there is an association b/w 2 variables.

Regression & Causation:-

- Causation means knowing whether x causes y to happen or vice versa.
- correlation & regression relationships are not the same as ~~caution~~ causation relationships.

Linearity & non-linearity Relationships:-

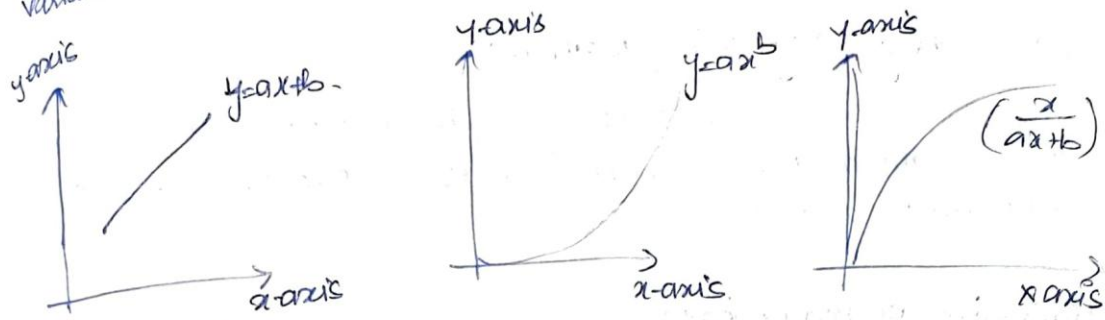
- The linearity relationship between the variables means the relationship between the dependent & independent variables, can be visualized as a straight line.

The line is of a form $y = mx + c$ can be fitted to the

$$y = ax + b$$

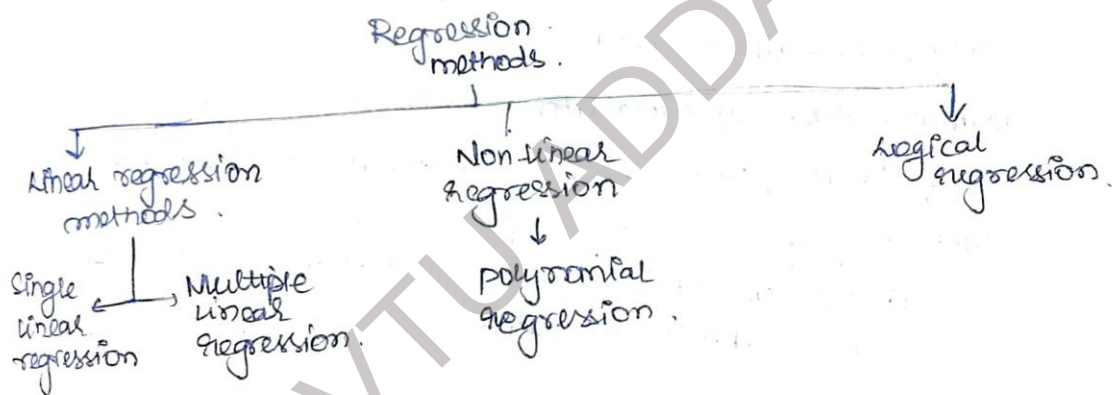
datapoints that indicate the relationship between independent & dependent variable.

The functions like exponential function $y = ax^b$ and power function $(y = \frac{x}{ax+b})$ are non-linear relationships b/w dependent variables that cannot be fitted in a line.



thus a linear relationship exist b/w x & y
 there also exist non-linear relationship b/w the variables such as an exponential function & a power function.

Types of Regression methods:-



Linear regression:-

It is a type of regression where a line is fitted upon a given data for finding the linear relationship between one independent variable and one dependent variable to describe relationships.

Multiple regression:-

It is a type of regression where a line is fitted for finding the linear relationships between 2 or more independent variables & one dependent variable to describe the relationships among variables.

Polynomial Regression:-

It is a type of non-linear regression method of describing relationships among variables where N^{th} is used to model the relationship between one dependent & one independent.

Logistic Regression:- is also called binary classifier.

- It is used for predicting categorical variables that involve one or more independent variables & 1 dependent variable.

Lasso & Ridge Regression methods:-

- These are special variants of regression methods where regularization methods are used to limit the number & size of coefficients of the independent variables.

Limitations of Regressions:-

- ① **Outliers** - Outliers are abnormal data. It can bias the outcome of the regression model, as outliers push the regression line towards it.
- ② **Number of cases** - The ratio of independent & dependent variables should be at least 20:1. For every explanatory variable there should be at least 20 samples.
- ③ **Missing data** - unfit model.
- ④ **multicollinearity** - If explanatory variables are highly correlated (0.9 & above), the regression is vulnerable to bias. Singularity leads to perfect correlation of 1.
 - ① The remedy is to remove explanatory variables that exhibit correlation more than 1.

Introduction to Linear Regression:-

- In the simplest form, the linear regression model can be created by finding the line among the scattered data points.

The line

$$y = a_0 + a_1 x + e \quad \text{--- (1)}$$

a_0 is intercept also known as bias.

a_1 is slope.

e is error factor in line that is getting fit } $\rightarrow$ regression coefficients
error in prediction.

The assumptions made:-

- The observations (y) are random and are mutually independent.
- The difference between the predicted value & true value is called error.

- The distribution of the error term is independent of the joint distribution of explanatory variables.
- the ~~the~~ unknown parameters of the regression models are constants.

$$e = y_i - (a_0 + a_1 x_i) \quad \text{--- (2)}$$

↑
error.

The value estimates of a_0 & a_1 are given as follows:

$$a_1 = \frac{(\sum xy) - (\sum x)(\sum y)}{(\sum x^2) - (\sum x)^2} \quad \text{--- (3)}$$

$$a_0 = (\bar{y}) - a_1 \bar{x} \quad \text{--- (4)}$$

Example:-

Let us consider an example where the five week's sales data (in thousands) is given as shown. Apply linear regression technique to predict 6th & 7th week sales.

x_i (week)	y_i (Sales)
1	1.2
2	1.8
3	2.6
4	3.2
5	3.8

Sol:-

x_i (week)	y_i (Sales)	$\bar{x} = \frac{1+2+3+4+5}{5} = 3$
1	1.2	$\bar{y} = \frac{1.2+1.8+2.6+3.2+3.8}{5} = 2.52$
2	1.8	$x^2 = 1, 4, 9, 16, 25$
3	2.6	$(\bar{x}^2) = \frac{1+4+9+16+25}{5} = 11$
4	3.2	$xy = (1 \times 1.2), (2 \times 1.8), (3 \times 2.6), (4 \times 3.2), (5 \times 3.8)$
5	3.8	$= 1.2, 3.6, 7.8, 12.8, 19$
		$(\sum xy) = \frac{1.2+3.6+7.8+12.8+19}{5} = 8.88$

$$a_1 = \frac{(\overline{xy}) - (\overline{x})(\overline{y})}{(\overline{x^2}) - (\overline{x})^2} = \frac{8.88 - 3 \times 2.52}{11 - 9}$$

$$a_1 = 0.66$$

$$a_0 = \overline{y} - a_1 \overline{x}$$

$$= 2.52 - 0.66 \times 3$$

$$a_0 = 0.54$$

$$\Rightarrow y = a_0 + a_1 x$$

$$y = 0.54 + 0.66x$$

7th week:

$$y = 0.54 + 0.66 \times 7 = 5.16$$

9th weeks

$$y = 0.54 + 0.66 \times 9 = 6.48$$

Linear Regression in Matrix form:-

The Matrix notations can be used for representing the values of independent and dependent variables.

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} + \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix}$$

This can be written as:

$y = Xa + e$ where X is an $n \times 2$ matrix,
 y is an $n \times 1$ vector,
 a is a 2×1 column vector,
 e is an $n \times 1$ column vector.

Example: Find linear regression of the data of week and product ~~total~~ sales.

x_i (weeks)	y_i (sales)
1	1
2	3
3	4
4	8

$$X \Rightarrow x' = [1 \ 2 \ 3 \ 4]$$

$$Y \Rightarrow y' = [1 \ 3 \ 4 \ 8]$$

in matrix form is:-

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}$$

$$Y = \begin{bmatrix} 1 \\ 3 \\ 4 \\ 8 \end{bmatrix}$$

The regression in matrix form is given as follows:-

$$a = (X^T X)^{-1} X^T Y$$

$$X^T X = \begin{bmatrix} 4 & 10 \\ 10 & 30 \end{bmatrix}$$

$$(X^T X)^{-1} = \begin{bmatrix} 1.5 & -0.5 \\ -0.5 & 0.2 \end{bmatrix}$$

$$(X^T X)^{-1} X^T Y = \begin{bmatrix} 1 & 0.5 & 0 & -0.5 \\ -0.3 & -0.1 & 0.1 & 0.3 \end{bmatrix}$$

$$((X^T X)^{-1} X^T Y) = \begin{bmatrix} -2 \\ 2.3 \end{bmatrix}$$

Multiple Linear Regression:-

Ex:- Apply multiple Regression for the value given when weekly sales along with sales for products.

X_1	X_2	Y
1	4	1
2	5	6
3	8	8
4	2	12

• Multiple regression model involves multiple predictors or independent variables

$$y = f(x_1, x_2) \quad a_0 + a_1x_1 + a_2x_2$$

~~$$y = f(x)$$~~

$$y = a_0 + a_1x_1 + a_2x_2 + \dots + a_nx_n + e$$

Soln:-

~~Given~~

Given:-

x_1	x_2	y
1	4	1
2	5	6
3	8	8
4	2	12

$$X = \begin{bmatrix} 1 & 1 & 4 \\ 1 & 2 & 5 \\ 1 & 3 & 8 \\ 1 & 4 & 2 \end{bmatrix}$$

$$y = \begin{bmatrix} 1 \\ 6 \\ 8 \\ 12 \end{bmatrix}$$

$$a = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix}$$

$$a = (X^T X)^{-1} X^T y$$

$$(X^T X) = \begin{bmatrix} 4 & 10 & 19 \\ 10 & 30 & 46 \\ 19 & 46 & 109 \end{bmatrix}$$

$$(X^T X)^{-1} = \begin{bmatrix} 3.153 & -0.59 & -0.3 \\ -0.59 & 0.2049 & 0.163 \\ -0.3 & 0.0943 & 0.0546 \end{bmatrix}$$

$$(X^T X)^{-1} X^T = \begin{bmatrix} 1.3606 & 0.4699 & -1.021 & 0.1912 \\ -0.319 & -0.098 & 0.1557 & 0.2622 \\ -0.065 & 5.4 \times 10^{-3} & 0.1857 & -0.125 \end{bmatrix}$$

$$a = \begin{bmatrix} -1.629 \\ 3.4834 \\ -0.054 \end{bmatrix} //$$

Polynomial Regression.

• If relationship between independent & dependent variable is not linear, linear regression cannot be used as results have large errors.

Problems can be solved by two methods:

- ① Transformation of non linear data to linear data, so that the linear regression can handle the data.
- ② using polynomial regression.

Transformations:

$y = ae^{bx}$ → exponential function transform into linear by applying $\log$ on both sides.

$$\ln y = bx + \ln a.$$

Also.

$$\log_{10} y = b \log_{10} x + \log_{10} a.$$

polynomial regression function

$$y = a_0 + a_1x + a_2x^2$$

3rd degree polynomial → cubic transformation.

$$y = a_0 + a_1x + a_2x^2 + a_3x^3.$$

Polynomials of maximum degree 4 are used, as high-order polynomials take some ~~strange~~ shape & makes curve more flexible.

error:-

$$E = \sum_{i=1}^n [y_i - (a_0 + a_1x_i + a_2x_i^2)]^2$$

error ~~max~~ minimized

Matrix form for polynomial regression:-

$$\begin{bmatrix} n & \sum x_i & \sum x_i^2 \\ \sum x_i & \sum x_i^2 & \sum x_i^3 \\ \sum x_i^2 & \sum x_i^3 & \sum x_i^4 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum (x_i y_i) \\ \sum (x_i^2 y_i) \end{bmatrix}$$

this is form $AX=B$ matrix can be represented of form

$$[a = X^{-1}B]$$

Eg: Consider the data provided in table and fit it using the second-order polynomial.

x	y
1	1
2	4
3	9
4	15

Sol:

x_i	y_i	x_i^2	x_i^3	x_i^4	$x_i y_i$	$x_i^2 y_i$
1	1	1	1	1	1	1
2	4	4	8	16	8	16
3	9	9	27	81	27	81
4	15	16	64	256	60	240
$\bar{x}=2.5$	$\bar{y}=7.25$	7.5				
		30	100	354	96	338

$$\begin{bmatrix} n & \sum x_i & \sum x_i^2 \\ \sum x_i & \sum x_i^2 & \sum x_i^3 \\ \sum x_i^2 & \sum x_i^3 & \sum x_i^4 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum (x_i y_i) \\ \sum (x_i^2 y_i) \end{bmatrix}$$

$n=4$

$$\begin{bmatrix} 4 & 10 & 30 \\ 10 & 30 & 100 \\ 30 & 100 & 354 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 29 \\ 96 \\ 338 \end{bmatrix}$$

$$\begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 29 \\ 96 \\ 338 \end{bmatrix} \begin{bmatrix} 4 & 10 & 30 \\ 10 & 30 & 100 \\ 30 & 100 & 354 \end{bmatrix}^{-1}$$

$$\begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} -0.75 \\ 0.95 \\ 0.75 \end{bmatrix}$$

$$y = -0.75 + 0.95x + 0.75x^2$$

Introduction to Decision Tree:-

Decision tree summarizes the information contained in the training dataset to the form of a tree structure.

Once the concept model is built, test data can be easily classified.

Structure of Decision Tree:-

consists

① Root node - at top.

② Internal decision nodes - tests input attributes leads to branching.

③ branches - gives labeled by list outcomes.

④ Leaf node - indicate classification labels.

• Each path from root to leaf represents a logical rules.

• The entire tree represents a set of classification rules in a structured form.

Nodes in Decision tree:-

○ → Root node.

◇ → Decision nodes.

□ → Leaf node.

Building The Tree:-

- Constructed in a top-down manner, starting from the root node.
- At each ~~tree~~ level the best split attribute is chosen.
- Recursively continues until only leaf nodes remain.
- The tree is complete when all paths lead to a leaf node.

Output:- Decision tree representing the hypothesis space.

Knowledge Inference / Classification:-

- Given a test instance, traverse the tree from the root.
- Evaluate test conditions at each decision node.
- Follow the corresponding branch until a leaf node is reached.

~~The~~

Advantage & Disadvantage of Decision Tree:-

Advantages:-

- ① Easy to model & interpret.
- ② Simple to understand.
- ③ Input & Output attribute can be discrete / continuous predictor variables.
- ④ Can model a high degree of non-linearity in relationship between the target variables & the predictor variables.
- ⑤ Quick to train.

Disadvantages:-

- ① Difficult to determine depth.

Entropy of Decision Tree:-

Entropy is a measure of uncertainty or randomness in data & is used in decision trees to determine the best attribute for splitting.

The best split feature provides the most information, ensuring accurate classification of test instances.

Entropy is calculated based on probability distributions.

Lower entropy indicates more homogenous data.

Entropy calculation by probability distribution.

Entropy of P is the information measure.

Entropy:

$$P = P_1 \dots P_n$$

$$\text{Entropy-Info}(P) = \text{Entropy-Info}(P_1 \dots P_n)$$

$$= -[P_1 \log_2(P_1) + P_2 \log_2(P_2) + \dots + P_n \log_2(P_n)]$$

Entropy-Info (6,4) is calculated as

$$= - \left[\frac{6}{10} \log_2 \frac{6}{10} + \frac{4}{10} \log_2 \frac{4}{10} \right]$$

$$\text{Entropy-Info}(X) = \sum_{\text{values}(X)} P_X[X=x] \log_2 \frac{1}{P_X[X=x]}$$

$P_X[X=x]$ probability of random variable x with a possible outcome.

$$\text{Note } \log_2 \frac{1}{P_X[X=x]} = -\log_2(P_X[X=x]).$$

General Algorithm for Decision Tree:-

1. Find the best attribute for the training dataset using an attribute selection measure and place it at the root of the tree.
2. Split the training dataset into subsets based on outcomes of the test attribute & each subset in a branch contains the data instances or tuples with the same value for the selected test attribute.
3. Repeat step 1 & step 2 on each subset until we end up in leaf nodes in all the branches of the tree.
4. This splitting process is recursive until the stopping criteria is reached.

Stopping Criterion

1. The data instances are homogeneous which means all belong to the same class C & hence its entropy is 0.
2. A node with some defined minimum number of data instances becomes a leaf.
3. The maximum tree depth is reached, so further splitting is not done and the node becomes a leaf node.

Entropy & Information Gain:

Entropy (Entropy-Info) measures the uncertainty in a dataset. It is calculated using:

$$\text{Entropy-Info}(T) = \sum_{i=1}^n P_i \log P_i \quad \text{--- (1)}$$

The entropy of an attribute A is given by:

$$\text{Entropy-Info}(T, A) = \sum_{i=1}^V \frac{|A_i|}{|T|} \times \text{Entropy-Info}(A_i) \quad \text{--- (2)}$$

where V is the number of distinct values in attribute A ,

$|A_i|$ is the number of instances for each distinct value.

$$\text{Information-Gain}(A) = \text{Entropy-Info}(T) - \text{Entropy-Info}(T, A) \quad \text{--- (3)}$$

Decision Tree Induction:-

Algorithm to construct a decision tree using ID3

- ① Compute the Entropy using eq. (1) for the entire training dataset using the target attribute.
- ② Compute Entropy using eq. (2) and information gain using eq. (3) for each attribute in the dataset.
- ③ Select the attribute with minimum entropy as the best split attribute.
- ④ Assign the best split attribute as the root node.
- ⑤ Now, split the root node into subsets based on the test condition forming subsets of the dataset.
- ⑥ Recursively apply the process to each subset until the leaf node is reached or no more training instances remain.

Note:-

- Branching stops if entropy is 0.
- At each iteration, the best split attribute is the one with the highest gain.

Example:-

Assess a student's performance during his course of study and predict whether a student will get a job offer or not in his final year of the course.

SNo	CGPA	Internships	Practical Knowledge	Communication Skills	Job offer
1.	≥ 9	Yes	Very good	Good	Yes
2.	≥ 8	No	Good	Moderate	Yes
3.	≥ 9	No	Average	Poor	No
4.	≥ 8	No	Average	Good	No
5.	≥ 8	Yes	Good	Moderate	Yes
6.	≥ 9	Yes	Good	Moderate	Yes
7.	< 8	Yes	Good	Poor	No
8.	≥ 9	No	Very good	Good	Yes
9.	≥ 8	Yes	Good	Good	Yes
10.	≥ 8	Yes	Average	Good	Yes

7 positive & 3 negative instances.

Formulas:

$$eq-①: Entropy-Info(T) = - \sum_{i=1}^n P_i \log P_i$$

$$eq-②: Entropy-Info(T, A) = \sum_{i=1}^n \frac{|A_i|}{|T|} \times Entropy-Info(A_i)$$

$$eq-③: Information-Gain(A) = Entropy-Info(T) - Entropy-Info(T, A)$$

S1: Calculate the ~~target~~ entropy of target class: Joboffer

$$Entropy-Info(T) = Entropy-Info(T, Joboffer)$$

$$Entropy-Info(T, Joboffer) = Entropy-Info(T, 1, 3)$$

$$= - \left[\frac{7}{10} \log_2 \frac{7}{10} + \frac{3}{10} \log_2 \frac{3}{10} \right]$$

$$Entropy-Info(Joboffer) = 0.88129$$

S2:

Iteration 1.

Calculate the entropy & Information Gain for each of the attribute in the given training dataset.

$$Entropy-Info(CGPA) = Entropy-Info(4, 4, 2)$$

$$\begin{matrix} \geq 9, \geq 8, < 8 \\ \downarrow \quad \downarrow \quad \downarrow \\ 4 \quad 4 \quad 2 \end{matrix}$$

CGPA	Joboffer = Yes	Joboffer = No	Total	Entropy
≥ 9	3	1	4	0.3245
≥ 8	4	0	4	0
< 8	0	2	2	0

$$Entropy-Info(Joboffer, CGPA(\geq 9)) = \sum_{i=1}^n \frac{|A_i|}{|T|} \times Entropy-Info(A_i)$$

indicates homogenous.

$$= \frac{4}{10} \left[\frac{3}{4} \log_2 \frac{3}{4} + \frac{1}{4} \log_2 \frac{1}{4} \right] + 0 + 0$$

$$= 0.3245$$

$$Gain = Entropy-Info(Joboffer) - Entropy-Info(Joboffer, CGPA)$$

$$= 0.88129 - 0.3245$$

$$Gain = 0.5568$$

A2
 $\text{Entropy-Info (Joboffer, Interactiveness)} = \sum_{i=1}^n \frac{|A_i|}{|T|} \times \text{Entropy-Info}(A_i)$

Interactiveness	Joboffer=Yes	Joboffer=No	Total	Entropy
Yes	5	1	6	0.39
No	2	2	4	0.40
				0.79001

Entropy-Info (Joboffer, Interactiveness)

$$= -\frac{6}{10} \left[\frac{5}{6} \log_2 \frac{5}{6} + \frac{1}{6} \log_2 \frac{1}{6} \right] - \frac{4}{10} \left[\frac{2}{4} \log_2 \frac{2}{4} + \frac{2}{4} \log_2 \frac{2}{4} \right]$$

= ~~0.39001~~ 0.79001

Gain = 0.88129 - 0.79001

A3
 Entropy-Info (Joboffer, PK) =

Gain = 0.09128

PK	Joboffer=Yes	Joboffer=No	Total	Entropy
Very good	2	0	2	0
good	4	1	5	0.36096
Average	1	2	3	0.2755 0.27548

Entropy-Info (Joboffer, PK)

$$= -\frac{5}{10} \left[\frac{4}{5} \log_2 \frac{4}{5} + \frac{1}{5} \log_2 \frac{1}{5} \right] - \frac{3}{10} \left[\frac{1}{3} \log_2 \frac{1}{3} + \frac{2}{3} \log_2 \frac{2}{3} \right]$$

= ~~0.63644~~ 0.63644

Gain = 0.88129 - 0.63644 = 0.24485 0.25485

A4
 Entropy-Info (Joboffer, CS)

CS	Joboffer=Yes	Joboffer=No	Total	Entropy
Good	4	1	5	0.36096
moderate	3	0	3	0
Poor	0	2	2	0

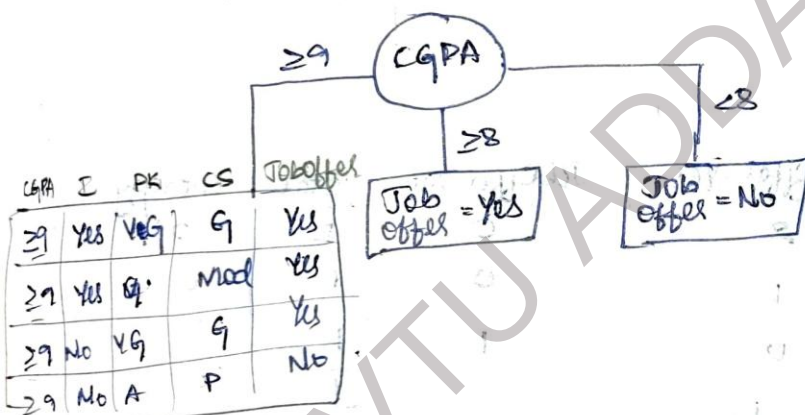
Entropy-Info (Joboffer, CS) =

$$= \frac{5}{10} \left[\frac{4}{5} \log_2 \frac{4}{5} + \frac{1}{5} \log_2 \frac{1}{5} \right] = \boxed{0.36096}$$

Gain = $0.88129 - 0.36096 = \boxed{0.52033}$

Attributes	Gain
CGPA	0.5568
Interactiveness	0.09128
Practical Knowledge	0.25485
Communication Skills	0.36 0.52033

max



CGPA	I	PK	CS	Joboffer
>= 9	Yes	Yes	G	Yes
>= 9	Yes	G	Med	Yes
>= 9	No	Yes	G	Yes
>= 9	No	A	P	No

Step 2

Iteration-2

S1: Entropy-Info (Joboffer) = Entropy-Info (3,1)

$$= - \left[\frac{3}{4} \log_2 \frac{3}{4} + \frac{1}{4} \log_2 \frac{1}{4} \right] = 0.81127$$

S2:

A2

Entropy-Info (Joboffer, I):

I	Joboffer=Yes	Joboffer=No	Total	Entropy
Yes	2	0	2	0
No	1	1	2	0.5
			<u>4</u>	<u>0.5</u>

$$\text{Entropy-Info (Joboffer, I)} = -\frac{2}{4} \left[\frac{1}{2} \log_2 \frac{1}{2} + \frac{1}{2} \log_2 \frac{1}{2} \right] + 0.$$

$$= 0.5$$

$$\text{Gain} = 0.81127 - 0.5 = 0.31127.$$

$$\boxed{\text{Gain} = 0.31127}$$

A3

PK	Joboffer=Yes	Joboffer=No	Total	Entropy
Very Good	2	0	2	0
Good	1	0	1	0
Average	0	1	1	0
			<u>4</u>	<u>0.</u>

$$\boxed{\text{Gain} = 0.81127}$$

A4

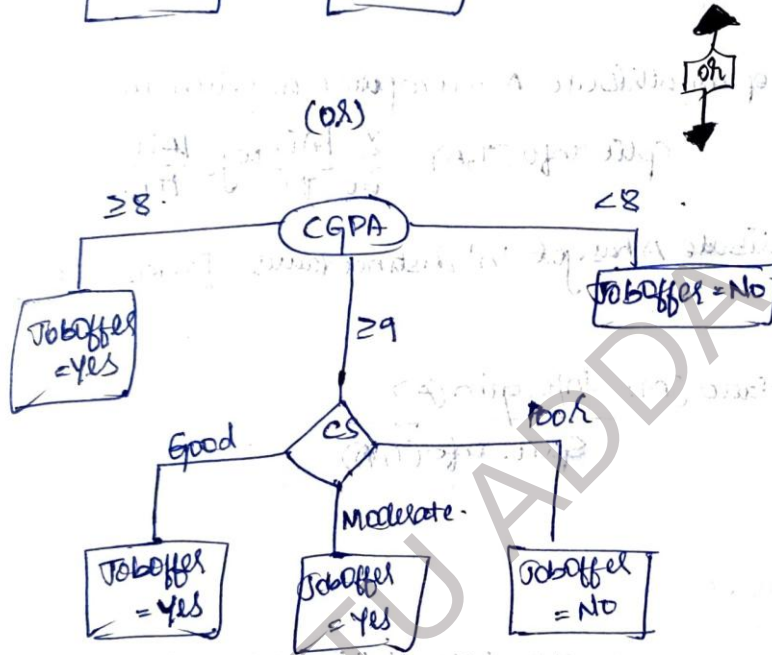
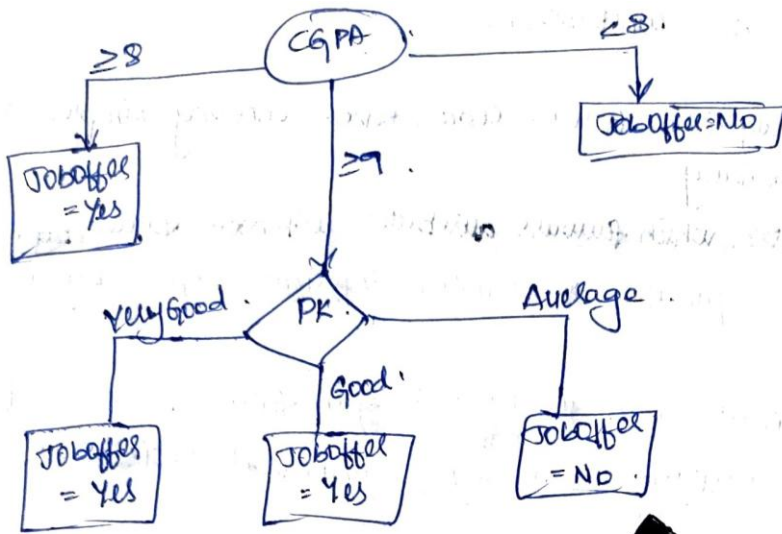
CS	Joboffer=Yes	Joboffer=No	Total	Entropy
Good	2	0	2	0
Med	1	0	1	0
Poor	0	1	1	0
			<u>4</u>	<u>0.</u>

$$\boxed{\text{Gain} = 0.81127}$$

Attribute	Gain
Interactiveness	0.31127
PK	0.81127
CS	0.81127

→ max

∴ Since Gain of both 'Practical knowledge' & 'Communication skills' are same, we select any one.



C4.5 Decision Tree algorithm:-

- C4.5 is an improvement over ID3 and handles both continuous and discrete attributes.
- It supports missing values and post pruning for better decision tree optimization.
- C5.0 is the successor of C4.5, offering more efficiency and generating smaller trees. Missing values are marked as '?' but are not included in calculations.

CH.5 Construction:-

- The CH.5 algorithm follows Occam's Razor, choosing simpler solutions for better accuracy.
- Unlike ID3, which favours attributes with more values, CH.5 uses Split-Ratio to normalize Information Gain by computing Split-Info.
- The attribute with the highest Split-Ratio is selected for splitting, ensuring a balanced & unbiased decision tree.

The Split-Info of an attribute A is computed as given in

$$\text{Split-Info}(T, A) = - \sum_{i=1}^n \frac{|A_i|}{|T|} \log_2 \frac{|A_i|}{|T|}$$

where, the attribute A has got 'n' distinct values $[A_1, A_2, \dots, A_n]$

$$\text{gain-ratio}(A) = \frac{\text{Info-Gain}(A)}{\text{Split-Info}(T, A)}$$

Algorithm for CH.5:-

- Compute Entropy-Info for the whole training dataset based on the target attribute.
- Compute Entropy-Info, Info-Gain, Split-Info and Gain-Ratio for each of the attribute in the training dataset.
- Choose the attribute for which Gain-Ratio is maximum as the best split attribute.
- The best split attribute is placed as the root node.
- The root node is branched into subtrees with each subtree as an outcome of the test condition of the root node attribute. Accordingly, the training dataset is also split into subsets.
- Recursively apply the same operation for the subset of the training set with the remaining attributes until a leaf node is defined or no more training instances are available in the subset.

eg:- Same as previous one.

Step 1: Calculate the entropy for whole training dataset.

$$\text{Entropy-Info (Joboffer)} = \text{Entropy-Info}(7, 3)$$

$$= - \left[\frac{7}{10} \log_2 \frac{7}{10} + \frac{3}{10} \log_2 \frac{3}{10} \right]$$

$$= 0.88129$$

Step 2:

Calculate

$$\text{Entropy-Info}(T, \text{CGPA}).$$

Attribute: CGPA.

CGPA	Joboffer=Yes	Joboffer=No	Total	Entropy
29	3	1	4	0.3245
28	4	0	4	0
28	0	2	2	0

$$\text{Entropy-Info}(T, \text{CGPA}) = - \frac{4}{10} \left[\frac{3}{4} \log_2 \frac{3}{4} + \frac{1}{4} \log_2 \frac{1}{4} \right]$$

$$= 0.32451$$

$$\boxed{\text{Entropy-Info} = 0.32451}$$

$$\text{Info-Gain} = \text{Entropy-Info}(T) - \text{Entropy-Info}(T, A)$$

$$= 0.88129 - 0.32451 = 0.55678$$

$$\boxed{\text{Info-Gain} = 0.55678}$$

$$\text{Split-Info} = - \sum_{i=0}^n \frac{|A_i|}{|T|} \log_2 \frac{|A_i|}{|T|}$$

$$= - \left[\frac{4}{10} \log_2 \frac{4}{10} + \frac{4}{10} \log_2 \frac{4}{10} + \frac{2}{10} \log_2 \frac{2}{10} \right]$$

$$= 1.521928$$

$$\boxed{\text{Split-Info} = 1.521928}$$

$$\text{Gain-Ratio (CGPA)} = \frac{\text{Info-Gain (CGPA)}}{\text{Split-Info (CGPA)}} = \frac{0.55678}{1.521928} = 0.36583$$

$$\boxed{\text{Gain-Ratio (CGPA)} = 0.36583}$$

→

Attribute: Interactiveness.

Interactiveness	Joboffer=Yes	Joboffer=No	Total	Entropy
Yes	5	1	6	0.39001
No	2	2	4	0.40
				0.79001

$$\text{Entropy-Info (T, Inter)} = 0.79001$$

$$\text{Info-Gain} = 0.8829 - 0.79001 = 0.09128$$

$$\text{Split-Info} = - \left[\frac{6}{10} \log_2 \frac{6}{10} + \frac{4}{10} \log_2 \frac{4}{10} \right]$$

$$= 0.97095$$

$$\text{Gain Ratio} = \frac{0.09128}{0.97095} = 0.094011$$

Attribute: Practical Knowledge.

PK	Joboffer=Yes	Joboffer=No	Total	Entropy
Very good	2	0	2	0
good	4	1	5	0.36096
Average	1	2	3	0.27548

$$\text{Entropy-Info} = 0.63644$$

$$\text{Info-Gain} = 0.8829 - 0.63644 = 0.25485$$

$$\text{Split-Info} = - \left[\frac{2}{10} \log_2 \frac{2}{10} + \frac{5}{10} \log_2 \frac{5}{10} + \frac{3}{10} \log_2 \frac{3}{10} \right]$$

$$= 1.48547$$

$$\text{Gain Ratio} = \frac{0.25485}{1.48547} = 0.171561$$

Attribute: Communication Skills.

	Yes	No	Total	Entropy
CS				0.36096
Good	4	1	5	
Moderate	3	0	3	0
Poor	3	2	5	

Entropy-Info = 0.36096.

Info-Gain = 0.52033.

$$\text{Split-Info} = - \left[\frac{5}{10} \log_2 \frac{5}{10} + \frac{3}{10} \log_2 \frac{3}{10} + \frac{2}{10} \log_2 \frac{2}{10} \right]$$

$$= 1.48547.$$

Gain-Ratio = $\frac{0.52033}{1.48547} = 0.35027.$

Step 3:

Attribute.

CGPA.

Interactiveness

PK.

CS.

Gain-Ratio.

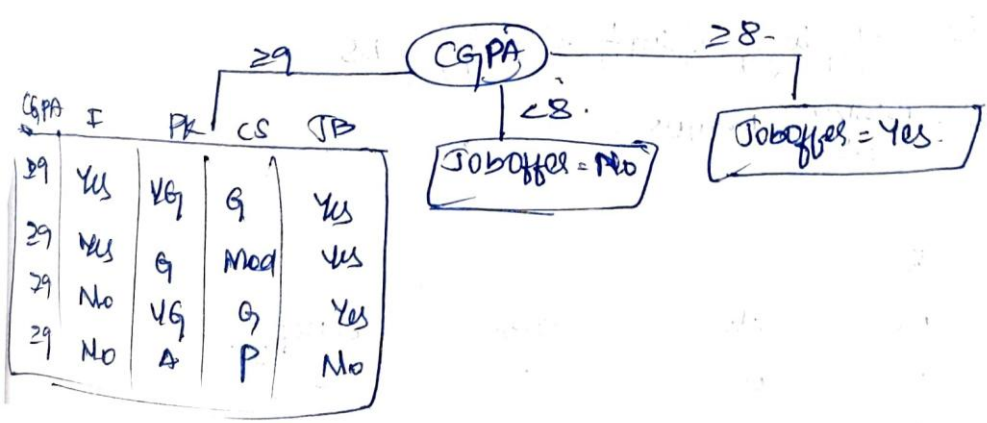
0.36583.

0.094011

0.171561

0.35027.

→ root, best split.
 → max.



Iteration-2

• equal probability
of samples.
Split info $\downarrow$
Entropy = 1

Entropy-Info (Robofter) = (3,1) = $-\left[\frac{3}{4} \log \frac{3}{4} + \frac{1}{4} \log \frac{1}{4}\right] = 0.81127$.

Attribute: Interactiveness -

	Yes	No	Total	Entropy:
Yes	2	0	2	0
No	1	1	2	0.5
			<u>4</u>	<u>0.5</u>

Entropy-Info (I, I) = 0.5.

Info-Gain = 0.31127.

Split-Info = $-\left[\frac{2}{4} \log \frac{2}{4} + \frac{2}{4} \log \frac{2}{4}\right] = 1$

Gain-Ratio = $\frac{0.31127}{1} = 0.31127$

Attribute: Practical Skills -

PK	Yes	No	Total	Entropy
Very good	2	0	2	0
Good	1	0	1	0
Average	0	1	1	0
			<u>4</u>	<u>0</u>

Entropy-Info = 0.

Info-Gain = 0.81127.

Split-Info = $-\left[\frac{2}{4} \log \frac{2}{4} + \frac{1}{4} \log \frac{1}{4} + \frac{1}{4} \log \frac{1}{4}\right] = 1.5$

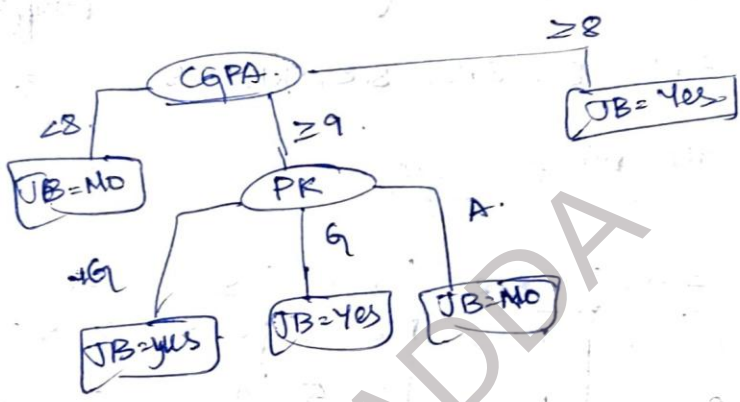
Gain-Ratio = $\frac{0.81127}{1.5} = 0.5408$.

Attribute: CR -

CR	Yes	No	Total	Entropy
Good	2	0	2	0
Med	1	0	1	0
Poor	0	1	1	0
			<u>4</u>	<u>0</u>

entropy-info = 0
 info-gain = 0.81127
 split-info = 1.5
 gain ratio = 0.5408

Attribute	Gain Ratio
Interactions	0.3127
PK	0.5408
CK	0.5408



q with continuous attributes in C4.5:-

Sho.	CGPA	Class
1	9.5	Yes
2	8.2	Yes
3	9.1	No
4	6.8	No
5	8.5	Yes
6	9.5	Yes
7	7.9	No
8	9.1	Yes
9	8.8	Yes
10	8.8	Yes

Ques 10

Step 1: Sort the given values in ascending order.

order.

6.8 No

7.9 No

8.2 Yes

8.5 Yes

8.8 Yes

8.8 Yes

9.1 No

9.1 Yes

9.5 Yes

9.5 Yes

Step 2: Remove duplicates after sorting.

$\Rightarrow$ 6.8, 7.9, 8.2, 8.5, 8.8, 9.1, 9.5

Total Entropy:

Entropy(7,3)

$$\Rightarrow -\left(\frac{7}{10} \log_2 \frac{7}{10} + \frac{3}{10} \log_2 \frac{3}{10}\right)$$

$$= 0.88129$$

Step 3: Calculate the gain for CGA.

	6.8		7.9		8.2		8.5		8.8		9.1		9.5	
Range	$\leq$	$>$	$\leq$	$>$	$\leq$	$>$	$\leq$	$>$	$\leq$	$>$	$\leq$	$>$	$\leq$	$>$
Yes	0	7	0	7	1	6	2	5	4	3	5	2	7	0
No	1	2	2	1	2	1	2	1	2	1	3	0	3	0
Entropy	0		0								0			0
	0.7642		0.5435		0.5916		0.65		0.812		0.88129			
					0.91829		1		0.91829		0.95443			

$$\text{Entropy of 6.8} \Rightarrow \frac{1}{10}(0) + \frac{9}{10}(0.7642) = 0.68778$$

$$7.9 \Rightarrow \frac{2}{10}(0) + \frac{8}{10}(0.5435) = 0.4348$$

$$8.2 \Rightarrow \frac{3}{10}(0.91829) + \frac{7}{10}(0.5916) = 0.689607$$

$$8.5 \Rightarrow \frac{4}{10}(1) + \frac{6}{10}(0.65) = 0.79$$

$$8.8 \Rightarrow \frac{6}{10}(0.91829) + \frac{4}{10}(0.812) = 0.875454$$

$$9.1 \Rightarrow \frac{8}{10}(0.95443) + \frac{2}{10}(0) = 0.763544$$

$$9.5 \Rightarrow \frac{10}{10}(0.88129) + \frac{0}{10}(0) = 0.88129$$

Step 4:- Identify the gain for all classes and choose the class with max gain.

Gain

Gain

$$\begin{aligned}
 6.5 &\Rightarrow 0.88129 - 0.68778 = 0.19351 \\
 7.9 &\Rightarrow 0.88129 - 0.4348 = 0.44649 \\
 8.2 &\Rightarrow 0.88129 - 0.68960 = 0.19169 \\
 8.5 &\Rightarrow 0.88129 - 0.79 = 0.09129 \\
 8.8 &\Rightarrow 0.88129 - 0.875454 = 0.00584 \\
 9.1 &\Rightarrow 0.88129 - 0.763544 = 0.117746 \\
 9.5 &\Rightarrow 0.88129 - 0.88129 = 0
 \end{aligned}$$

max gain $\rightarrow$ class 7.9

tips:

Hence, CGPA 7.9 is chosen as the splitting point, we can discretize CGPA into 2 categories.

- ① $CGPA \leq 7.9$
- ② $CGPA > 7.9$

Discretized Instances:-

<u>Sno.</u>	<u>CGPA</u>	<u>CGPA Discretized</u>	<u>Joboffer</u>
1	9.5	> 7.9	Yes
2	8.2	> 7.9	Yes
3	9.1	> 7.9	No
4	6.8	≤ 7.9	No
5	8.5	> 7.9	Yes
6	9.5	> 7.9	Yes
7	7.9	≤ 7.9	No
8	9.1	> 7.9	Yes
9	8.8	> 7.9	Yes
10	8.8	> 7.9	Yes

CART

- CART stands for Classification & Regression Trees Construction.
- CART is a multivariate decision tree - algorithm.
- It can be used for both continuous and categorical valued variables.
- It solves both classification & regression problems.
- Gini-Index is used to construct decision tree in CART.

$$\text{Gini-Index}(S) = 1 - \sum_{i=1}^m P_i^2$$

$$\text{Gini-Index}(T, A) = \frac{|S_1|}{|T|} \text{Gini}(S_1) + \frac{|S_2|}{|T|} \text{Gini}(S_2)$$

Steps:-

- Compute Gini-Index for the whole training dataset based on the target attribute.
- Compute Gini-Index for each of the attribute and for the subsets of each attribute in the training dataset.
- Choose the best splitting subset which has minimum Gini-Index for an attribute.
- "
- "
- "
- "
- "

Example:-

• Sample is same as ID3 sample: Joboffer ? or X sample.

Aim: Construct decision tree using CART algorithm for the data samples given ~~steps~~ in ID3 algorithm.

Step 1: Calculate Gini Index for complete data samples 10 of which 7 are positive & 3 are negative.

$$\text{Gini-index}(S) = 1 - \left[\left(\frac{7}{10} \right)^2 + \left(\frac{3}{10} \right)^2 \right]$$

$$\boxed{\text{Gini-index}(T) = 0.42 //}$$

Step 2: Calculate Gini-Index for each of the attribute with respect to the subsets in the attribute.

Attribute - CGPA

CGPA	Yes	No
≥ 9	3	1
≥ 8	4	0
< 8	0	2

$$\Rightarrow \{ \geq 9, \geq 8 \} + \{ < 8 \}$$

$$\Rightarrow \{ \geq 9, < 8 \} + \{ \geq 8 \}$$

$$\Rightarrow \{ \geq 8, < 8 \} + \{ \geq 9 \}$$

$$\text{Gini-index } (T, \text{CGPA} \in \{ \geq 9, \geq 8 \}) = 1 - \left[\left(\frac{7}{8} \right)^2 + \left(\frac{1}{8} \right)^2 \right] = 0.21875$$

$$\text{Gini-index } (T, \text{CGPA} \in \{ < 8 \}) = 1 - \left[\left(\frac{0}{2} \right)^2 + \left(\frac{2}{2} \right)^2 \right] = 0 //$$

$$\text{Gini-index } (T, \text{CGPA} \in \{ \geq 9, \geq 8 \}, \{ < 8 \}) = \frac{8}{10} (0.21875) + \frac{2}{10} (0) = 0.175$$

To compute Gini-index of subset attributes. $\frac{|S_1|}{|T|} \text{Gini}(S_1) + \frac{|S_2|}{|T|} \text{Gini}(S_2)$

$$\text{Gini-index } (T, \text{CGPA} \in \{ \geq 9, < 8 \}) = 1 - \left[\left(\frac{3}{6} \right)^2 + \left(\frac{3}{6} \right)^2 \right] = 0.5$$

$$\text{Gini-index } (T, \text{CGPA} \in \{ \geq 8 \}) = 1 - \left[\left(\frac{4}{4} \right)^2 + \left(\frac{0}{4} \right)^2 \right] = 0$$

$$\text{Gini-index } (T, \text{CGPA} \in \{ \geq 9, < 8 \}, \{ \geq 8 \}) = \frac{6}{10} \text{Gini}(S_1) + \frac{4}{10} \text{Gini}(S_2) = 0.3$$

$$\text{Gini-index } (T, \text{CGPA} \in \{ \geq 8, < 8 \}) = 1 - \left[\left(\frac{4}{6} \right)^2 + \left(\frac{2}{6} \right)^2 \right] = 0.444$$

$$\text{Gini-index } (T, \text{CGPA} \in \{ \geq 9 \}) = 1 - \left[\left(\frac{3}{4} \right)^2 + \left(\frac{1}{4} \right)^2 \right] = 0.375$$

$$\text{Gini-index } (T, \text{CGPA} \in \{ \geq 8, \geq 9 \}) = \frac{6}{10} \text{Gini}(S_1) + \frac{4}{10} \text{Gini}(S_2) = 0.4164$$

Attribute - CGPA

Subsets			Gini-index
≥ 9	≥ 8	< 8	0.175
≥ 9	< 8	≥ 8	0.3
≥ 8	< 8	≥ 9	0.4164

Choose the best splitting subset with minimum Gini-index.

∴ Subset:- $\{ \geq 9, \geq 8 \}, \{ < 8 \}$

S4: Calculate $Gini$ (The best splitting subset of the given attribute).

$$\underline{Gini}(CQPA) = Gini(T) - Gini(T, CQPA)$$

$$= 0.42 - 0.175$$

$$\boxed{Gini = 0.245}$$

Attribute: Interactiveness

Interactiveness	Yes	No
Yes	5	1
No	2	2

Subsets:

{Yes, No}

↓
{Yes} {No}

$$Gini_index(T, It \in \{Yes\}) = 1 - \left[\left(\frac{5}{6} \right)^2 + \left(\frac{1}{6} \right)^2 \right] = 0.278$$

$$Gini_index(T, It \in \{No\}) = 1 - \left[\left(\frac{2}{4} \right)^2 + \left(\frac{2}{4} \right)^2 \right] = 0.5$$

$$Gini_index(T, Interactiveness) = \frac{6}{10} (Gini(S_1)) + \frac{4}{10} (Gini(S_2))$$

$$= 0.3668$$

$$Gini(Interactiveness) = Gini(T) - Gini(T, It)$$

$$= 0.42 - 0.3668$$

$$\boxed{Gini = 0.0532}$$

Attribute: Practical ~~know~~ knowledge

PK	Yes	No
VG	2	0
G	4	1
A	1	2

Subsets:

{VG, G}, {A}

{VG, A}, {G}

{G, A}, {VG}

$$Gini_index(T, PK \in \{VG, G\}) = 1 - \left[\left(\frac{6}{7} \right)^2 + \left(\frac{1}{7} \right)^2 \right] = 0.24189$$

$$Gini_index(T, PK \in \{A\}) = 1 - \left[\left(\frac{1}{3} \right)^2 + \left(\frac{2}{3} \right)^2 \right] = 0.4444$$

$$Gini_index(T, PK \in \{VG, G\}, \{A\}) = \frac{7}{10} \times (0.24189) + \frac{3}{10} \times (0.4444)$$

$$= 0.304743$$

$$\text{Gini-index}(T, PK \in \{VG, A\}) = 1 - \left[\left(\frac{2}{5} \right)^2 + \left(\frac{3}{5} \right)^2 \right] = 0.48$$

$$\text{Gini-index}(T, PK \in \{G\}) = 1 - \left[\left(\frac{4}{5} \right)^2 + \left(\frac{1}{5} \right)^2 \right] = 0.32$$

$$\text{Gini-index}(T, PK \in \{VG, A\}, \{G\}) = \frac{5}{10} \times 0.48 + \frac{5}{10} \times 0.32 = 0.4$$

$$\text{Gini-index}(T, PK \in \{G, A\}) = 1 - \left[\left(\frac{5}{8} \right)^2 + \left(\frac{3}{8} \right)^2 \right] = 0.46875$$

$$\text{Gini-index}(T, PK \in \{VG\}) = 1 - \left[\left(\frac{2}{2} \right)^2 + \left(\frac{0}{2} \right)^2 \right] = 0$$

$$\text{Gini-index}(T, PK \in \{G, A\}, \{VG\}) = \frac{8}{10} \times 0.46875 + \frac{2}{10} \times 0 = 0.375$$

Subsets			Gini-index
VG	G	A	0.304743
VG	A	G	0.4
G	A	VG	0.375

$\therefore$ Best split Subset $\Rightarrow \{VG, G\}, \{A\}$ with Gini-index 0.304743.

$$\text{Gini}(PK) = \text{Gini}(T) - \text{Gini}(T, PK)$$

$$= 0.42 - 0.304743$$

$$\boxed{\text{Gini}(PK) = 0.115257}$$

Attribute: Computer Skills.

CS	Yes	No
Good	4	1
Moderate	3	0
Poor	0	2

Subsets

$\{G, M\}, \{P\}$

$\{G, P\}, \{M\}$

$\{P, M\}, \{G\}$

$$\text{Gini-index}(T, CS \in \{G, M\}) = 1 - \left[\left(\frac{7}{8} \right)^2 + \left(\frac{1}{8} \right)^2 \right] = 0.21875$$

$$\text{Gini-index}(T, CS \in \{P\}) = 1 - \left[\left(\frac{0}{2} \right)^2 + \left(\frac{2}{2} \right)^2 \right] = 0$$

$$\text{Gini-index}(T, CS \in \{G, M\}, \{P\}) = \frac{8}{10} \times 0.21875 + \frac{2}{10} \times 0 = 0.175$$

$$\text{Gini-index}(T, CS \in \{G, P\}) = 1 - \left[\left(\frac{4}{7} \right)^2 + \left(\frac{3}{7} \right)^2 \right] = 0.48979$$

$$\text{Gini-index}(T, CS \in \{M\}) = 1 - \left[\left(\frac{3}{3} \right)^2 + \left(\frac{0}{3} \right)^2 \right] = 0$$

$$\text{Gini-index}(T, CS \in \{G, P\}, \{M\}) = \frac{7}{10} \times 0.48979 + \frac{3}{10} \times 0$$

$$= 0.342853$$

$$\text{Gini-index}(T, CS \in \{P, M\}) = 1 - \left[\left(\frac{3}{5} \right)^2 + \left(\frac{2}{5} \right)^2 \right] = 0.48$$

$$\text{Gini-index}(T, CS \in \{G\}) = 1 - \left[\left(\frac{4}{5} \right)^2 + \left(\frac{1}{5} \right)^2 \right] = 0.32$$

$$\text{Gini-index}(T, CS \in \{P, M\}, \{G\}) = \frac{5}{10} \times 0.48 + \frac{5}{10} \times 0.32$$

$$= 0.4$$

Subsets			Gini-index
G	M	P	0.175
G	P	M	0.3428
M	P	G	0.4

$\therefore$ Best splitting subset is $\{G, M\}, \{P\}$ with Gini-index as 0.175

$$\text{Gini}(CS) = \text{Gini}(T) - \text{Gini}(T, CS)$$

$$= 0.42 - 0.175$$

$$\boxed{\text{Gini}(CS) = 0.245}$$

Attributes	Gini
CGPA	0.2455
Interactiveness	0.0532
PK	0.11525
CS	0.245

• Maximum Gini $\Rightarrow$ Best splitting and root = CGPA

Decision Tree -



≥ 9	Yes	VG	G	Yes
≥ 8	No	G	M	Yes
≥ 9	No	AV	P	No
≥ 8	Yes	G	M	Yes
≥ 9	Yes	G	M	Yes
≥ 9	No	VG	G	Yes
≥ 8	Yes	G	G	Yes
≥ 8	Yes	A	G	Yes

Iteration - 2 -

Dataset -

GPA	Intellectiveness	PK	CS	JB
≥9	Yes	NG	G	Yes
≥8	No	G	M	Yes
≥9	No	A	P	No
≥8	Yes	G	M	Yes
≥9	Yes	G	M	Yes
≥9	No	VG	G	Yes
≥9	Yes	G	G	Yes
≥8	Yes	G	G	Yes
≥8	Yes	A	G	Yes

$$G_{in}(T) = 1 - \left[\left(\frac{7}{8} \right)^2 + \left(\frac{1}{8} \right)^2 \right]$$

Attribute: Intellectiveness

I	Yes	No
Yes	5	0
No	2	1

$$G_{in_index}(I, I \in \{Yes\}) = 1 - \left[\left(\frac{5}{5} \right)^2 + \left(\frac{0}{5} \right)^2 \right]$$

$$G_{in_index}(T, I \in \{No\}) = 1 - \left[\left(\frac{2}{3} \right)^2 + \left(\frac{1}{3} \right)^2 \right]$$

VTU ADDA